IDENTITY AND ACCESS MANAGEMENT (IAM) IN AWS

IAM (IDENTITY AND ACCESS MANAGEMENT)

It refers to a framework or policies and technologies for ensuring that the proper people in an organisation have the appropriate access to the technology resources.

1. AWS Identity and Access Management (IAM) is a web service that provides secure access to AWS resources.
2. we can use IAM to control who is authenticated (signed-in) and authorised (has permission) to use resources.
3. It allows you to manage users, groups, and permissions to control who can access which resources in your AWS account.
4. when we create a account you begin with a single sign-in identity that has completely access to all aws services and resources in account.
5. this identity is called aws account "root user" and accessed by sign-in with the email address or password.
6. but aws recommend that you don't use root user for everyday task even the administrative once's.
7. use other IAM account to manage the administrator task of your account and secured lock away the root user credential and use them to perform only a few account and service management task.

AWS Identity and Access Management is a web service that you securely control access to AWS resources. We use IAM to control who is authenticated (signed-in) and authorized (has permission) to use resources.

IAM USER

1. IAM User is an identity in Amazon Web Services that represents a person or application needing access to AWS.
2. Each IAM User has unique credentials like a username/password (for console) or access key & secret key (for API/CLI access).
3. By default, a new IAM User has no permissions, and you must assign permissions using IAM policies or groups.
4. IAM Users belong to a single AWS account, and all usage is billed to the parent account, not the user.
5. By default, we can create up to 5000 users per aws account.
6. And create upto 10 users at a time.

IAM GROUPS

1. IAM Group is a collection of IAM Users in Amazon Web Services used to manage multiple users together.
2. an IAM group is an identity in IAM that specify collection of IAM user.
3. we can add no of users in a group so that we can specify permission for multiple users at a same time.
4. group make permission easier to manage for large set of users.
5. By default, we can create upto 300 groups in an AWS account.
6. And a user can be a member of up to 10 IAM groups.

IAM ROLE

1. IAM role is an identity in IAM that we create in our own account that have the specific permission.
2. IAM role in aws works like a IAM user.
3. IAM Role is an identity in Amazon Web Services with specific permissions, similar to an IAM user but used differently.
4. IAM Role does not have permanent credentials (no password or access key); it provides temporary security credentials when assumed.
5. Roles are assumed by users, applications, or services to perform actions (e.g., an EC2 instance accessing S3).
6. IAM Roles are used for secure access scenarios like cross-account access, AWS services (service roles), or external users (federation/web identity).
7. By default we can create upto 1000 IAM roles.

TYPES OF IAM ROLES

1. **AWS Service Role**
   Used by AWS services (like EC2, Lambda) to access other AWS resources.

2. **SAML 2.0 federation**
   Used for enterprise login (SSO) via identity providers like Active Directory.

3. **AWS Cross-Account Role**
   Allows one AWS account to access resources in another AWS account.

4. **Web Identity Role**
   Used for external users via providers like Google, Facebook (OIDC).

5. **Custom trust policy**

POLICIES

1. An policy is an object in AWS where is associate with an entity or resources define their permission.
2. the permission in the policies that determines whether the request is allowed or denied.
3. most policies are stored in AWS as JSON documents.
4. a policy defines the permissions to the IAM identity.

TYPES OF POLICIES (Where they are used)

1. **Identity Based Policies**
    1. Attached to users, groups, or roles
    2. Define what actions they can perform

2. **Resources Based Policies**
    1. Attached directly to a resource (like S3 bucket)
    2. Define who can access that resource

3. **Permission Boundary**
    1. Sets the maximum permissions limit for a user or role
    2. Even if more permissions are given, they cannot exceed this limit

4. **Session Policy**
    1. Temporary policies applied during a role session
    2. Further restrict permissions for that session

5. **ACLs (Access Control List)**
    1. Used mainly in services like S3
    2. Define basic permissions (read/write) at resource level

6. **SCPs (Service Control Policies)**
    1. Used in AWS Organizations
    2. Control permissions across multiple AWS accounts

WHAT ARE THE WAYS TO ATTACHED POLICY TO A USER

1. add user into the group
2. copy permissions of existing user
3. attach directly

TYPES OF POLICIES (Who created them)

Aws Managed Policies

1. AWS managed policies are predefined policies created and maintained by AWS.
2. They are designed to provide common permissions needed for specific AWS services or tasks
3. These policies are accessible across multiple AWS accounts and regions.
4. AWS managed policies cannot be edited, and their permissions are managed by AWS.
5. You can directly attach AWS managed policies to IAM identities (users, groups, or roles) or use them as a reference in a custom policy.

Customer Policy

1. Custom managed policies are policies that you create and manage yourself.
2. They offer flexibility in defining fine-grained permissions tailored to your specific requirements.
3. Custom managed policies are attached to IAM identities (users, groups, or roles) to grant permissions.
4. They can be edited, updated, and version-controlled according to your needs.
5. custom managed policies require manual updates to incorporate new services or changes.

Inlined policy

1. Inlined policies are useful for granting specific permissions to an individual user, group, or role without the need to create a separate managed policy.
2. Inlined policies are written in JSON format and provide explicit permissions for the associated entity.
3. An inlined policy is an IAM policy that is directly attached to an IAM user, group, or role.
4. Inlined policies are defined and managed within the IAM entity to which they are attached.

BOUNDARY PERMISSION

1. Boundary permissions, also known as a boundary policy, are used to set the maximum permissions that can be granted to an IAM entity (user, group, or role).
2. A boundary permission is an IAM policy that is attached to an IAM entity.
3. When a boundary policy is assigned to an IAM entity, it restricts the permissions that can be granted to that entity through other means, such as inlined policies or managed policies.

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5. Boundary policies are useful for enforcing least privilege principles and preventing excessive permissions from being granted to IAM entities.

HOW TO CREATE A USER

1. Click Users → Create user
2. Enter username -----> mayur
3. Select access type (Console / Programmatic)
4. Attach permissions (group or policy)
5. Click Create user
6. Download credentials

HOW TO CREATE A GROUP

1. Click User groups → Create group
2. Enter Group name ----> tech
3. Attach permissions (policies)
4. Add users (optional)
5. Click Create group

HOW TO CREATE A ROLE

1. Got to IAM service
2. Click Roles → Create role
3. Select AWS service
4. Choose EC2
5. Attach permissions (policies) -----> AdministratorAccess
6. Enter Role name -----> Role-For-EC2
7. Create role

HOW TO CREATE A INLINED POLICY

VISUAL FORMAT

1. click on user
2. click on add permissions
3. create a inlined policy
4. choose policy type – Jason/visual/

5. select service – s3
6. Actions → Select (e.g., GetObject, PutObject)
7. Resources → Choose specific bucket or all
8. policy name → my-inlined-policy
9. create a policy

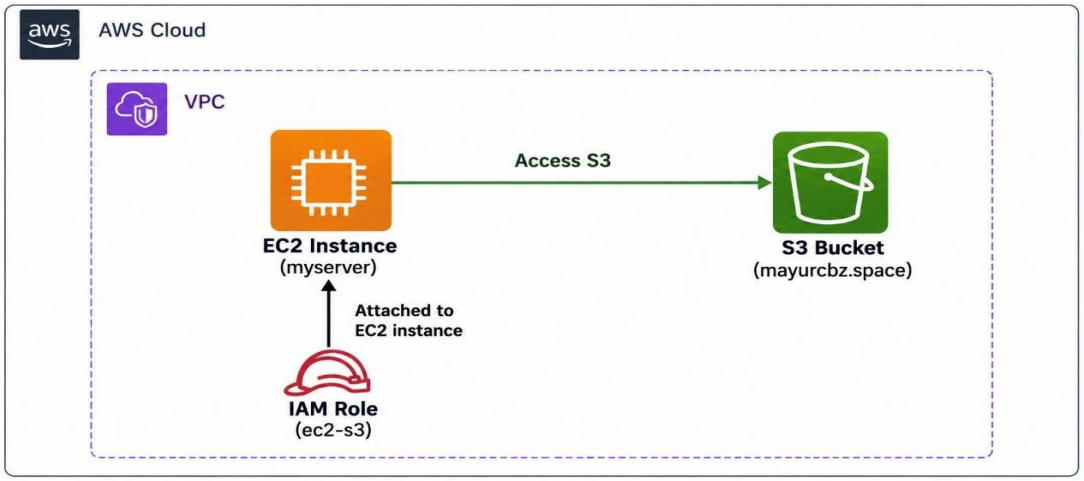
IN JASON FORMAT

```
{
 "Version": "2012-10-17",
 "Statement": [
  {
   "Effect": "Allow",
   "Action": "s3:ListBucket",
   "Resource": "*"
  }
 ]
}
```

Note:

1. This policy is directly attached to a user so if the user is deleted the policy is also deleted
2. This policy we can attach to a user / group / role

ACCESS S3 USING IAM ROLE



create a server

1. name - myserver
2. instance_type - t3.micro
3. key_name - mayur.pem
4. sg - mysg
5. create a server

create a s3 bucket

1. name - mayurcbz.space

create a Iam role (ec2-s3)

1. search and click on IAM service
2. create a role
3. choose - aws-service
4. permission - s3fullAccess
5. create a role

Assign role to server

1. click on server
2. click on action
3. click on security
4. click on modify iam role
5. select-role - (ec2-s3)
6. update

Access the server

1. apt update
2. snap install aws-cli --classic
3. aws s3 ls
